

Table 3. Proportions of diet composition for the western North Pacific Ecopath model. Functional groups (numbered) were defined in coastal Oyashio (OYC), coastal Kuroshio (KC), and offshore (OF) regions (in parentheses). Values in **bold** indicate adjusted diet composition to achieve thermodynamic balancing

No	Group name	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	
1	Baleen whales (OYC & OF)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
2	Toothed whales (all)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
3	Seabirds (all)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
4	Sharks (all)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
5	Tunas (all)	0	0	0	0	0.001	0.001	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
6	Skipjack (all)	0	0	0	0	0.001	0.001	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
7	Miscellaneous piscivores (all)	0	0	0	0	0.09	0.09	0.001	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
8	Yellowtail (OYC & KC)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
9	Japanese sardine (all)	0.01	0.01	0.10	0.06	0.09	0.05	0.04	0.05	0	0	0	0.01	0.01	0	0.01	0	0	0	0	0	0	0.06	0	0	0	0	0.01	0	0	0	0	0	0	0	0	0
10	Japanese anchovy (all)	0.05	0.05	0.10	0.06	0.09	0.05	0.04	0.11	0	0	0	0.24	0.24	0	0.19	0	0	0	0	0	0	0.06	0	0	0	0	0.01	0	0	0	0	0	0	0	0	0
11	Pacific saury (all)	0.02	0.02	0.10	0.06	0.09	0.089	0.07	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.01	0	0	0	0	0	0	0	0	0
12	Chub mackerel (all)	0.01	0.011	0.10	0.06	0.09	0.089	0.08	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.01	0	0	0	0	0	0	0	0	0
13	Spotted mackerel (all)	0.01	0.011	0.10	0.06	0.09	0.089	0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.01	0	0	0	0	0	0	0	0	0
14	Round herring (KC)	0	0	0.004	0.005	0.005	0.005	0.005	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.005	0	0	0	0	0.01	0	0	0	0	0	0	0	0	0
15	Jack mackerel (OYC & KC)	0.01	0.001	0.003	0.005	0.001	0.001	0.005	0.005	0	0	0	0	0	0	0	0	0	0	0	0	0	0.005	0	0	0	0	0.005	0	0	0	0	0	0	0	0	0
16	Righteye flounders (OYC)	0	0	0	0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0.05	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
17	Walleye pollock (OYC)	0	0	0	0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
18	Pacific cod (OYC)	0	0	0	0.058	0	0	0.01	0.05	0	0	0	0	0	0	0	0	0	0	0	0	0	0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0
19	Miscellaneous bottom fishes (OYC)	0	0	0	0.06	0	0	0.02	0.09	0	0	0	0	0	0	0	0.20	0	0.10	0	0	0	0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	Flatfishes (KC)	0	0	0	0.01	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
21	Seabreams (KC)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
22	Demersal piscivores (KC)	0	0	0	0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.05	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
23	Miscellaneous bottom fishes (KC)	0	0	0	0.06	0	0	0.069	0.08	0	0	0	0	0	0	0	0	0	0	0	0	0.20	0.20	0.11	0	0	0	0	0	0	0	0	0	0	0	0	
24	Mesopelagic fishes (OYC)	0	0.10	0.10	0.064	0.091	0.12	0.12	0.10	0	0	0	0	0	0	0	0.10	0.10	0.21	0	0	0	0	0	0	0	0	0.04	0.05	0	0	0	0	0	0	0	0
25	Mesopelagic fishes (KC)	0	0.10	0.10	0.064	0.085	0.11	0.12	0.10	0	0	0	0	0	0	0	0	0	0	0	0	0.10	0.10	0.11	0	0	0	0	0.04	0.05	0	0	0	0	0	0	0
26	Mesopelagic fishes (OF)	0	0.19	0.10	0.064	0.091	0.12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.04	0.05	0	0	0	0	0	0	0	0
27	Epipelagic cephalopods (all)	0.01	0.20	0.10	0.064	0.091	0.091	0.09	0.105	0	0	0	0.01	0.01	0	0.10	0.10	0	0.11	0	0.10	0.10	0.06	0	0	0	0	0.03	0.05	0	0	0	0	0	0	0	0
28	Mesopelagic cephalopods (all)	0	0.20	0.10	0.064	0.091	0.091	0.09	0.11	0	0	0	0.01	0.01	0	0.10	0.10	0.10	0.21	0	0.10	0.10	0.06	0	0	0	0	0.03	0.05	0	0	0	0	0	0	0	0
29	Benthos (OYC)	0	0.051	0	0	0	0	0.09	0.10	0	0	0	0	0	0	0	0.20	0.30	0.21	0.30	0	0	0	0	0.25	0	0	0	0	0.10	0	0	0	0	0	0	
30	Benthos (KC)	0	0.051	0	0	0	0	0.09	0.10	0	0	0	0	0	0	0	0	0	0	0	0	0.30	0.45	0.41	0.50	0	0.25	0	0	0	0	0	0.10	0	0	0	
31	Krill (OYC)	0.10	0	0.001	0.001	0.001	0.001	0	0	0.10	0.10	0.10	0.10	0.10	0	0.20	0.20	0.30	0.11	0.30	0	0	0	0	0	0.50	0	0	0.20	0.20	0	0	0	0	0	0	0
32	Krill (OF)	0.50	0	0.001	0.001	0.001	0.001	0	0	0.20	0.10	0.10	0.20	0.20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.19	0.20	0.20	0	0	0	0	0	0
33	Zooplankton (OYC)	0.14	0	0.001	0	0.001	0.001	0	0	0.10	0.15	0.15	0.10	0.10	0	0.20	0.10	0.20	0	0.40	0	0	0	0	0	0.25	0	0	0.12	0.15	0.10	0	0.10	0	0	0	0
34	Zooplankton (KC)	0	0	0.001	0	0.001	0.001	0	0	0.10	0.15	0.15	0.10	0.10	1.00	0.20	0	0	0	0	0	0.20	0	0	0.50	0	0.75	0	0.115	0.05	0	0.10	0	0	0	0	0
35	Zooplankton (OF)	0.14	0	0.001	0	0.001	0.001	0	0	0.35	0.50	0.50	0.23	0.23	0	0	0	0	0	0	0	0	0	0	0	0	0	0.81	0.12	0.15	0	0	0	0.10	0	0	0
36	Phytoplankton (OYC)	0	0	0	0	0	0	0	0.05	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.10	0	0.70	0	1.00	0	0	0
37	Phytoplankton (KC)	0	0	0	0	0	0	0	0.05	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.70	0	1.00	0	0
38	Phytoplankton (OF)	0	0	0	0	0	0	0	0.05	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.70	0	1.00	0
39	Detritus (OYC)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.70	0	0.20	0	0	0	0	
40	Detritus (KC)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.70	0	0	0	0	
41	Detritus (OF)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.20	0	0	0	